

PYTHON PROGRAMMING UNITS

Applicable To:

Data Analytics

Data Science

Machine Learning Engineering

Data Engineering

Front-End Development

Full-Stack Development

Back-End Development

Welcome to the World of Python

100 mins

Python is a high-level, interpreted, general-purpose programming language. It is designed to be easy to learn and use, with a syntax that is clear and concise. Python is also a dynamic language, meaning that it can adapt to different environments and data types.

Key Topics:

- Python's history and evolution
- Python's syntax and semantics
- Python's data types and variables
- Python's control flow statements
- Python's functions and modules
- Python's error handling and debugging

Control Flow and Conditional Logic

100 mins

Control flow is the sequence of instructions that a program executes. Conditional logic allows a program to make decisions based on certain conditions and execute different code blocks based on the outcome of those decisions.

Key Topics:

- Conditional statements (if, elif, else)
- Loops (for, while)
- Break and continue statements
- Pass statement

Data Structures - Lists, Tuples, and Dictionaries

100 mins

Data structures are ways of organizing and storing data. Lists, tuples, and dictionaries are some of the most commonly used data structures in Python.

Key Topics:

- Lists (mutable, ordered)
- Tuples (immutable, ordered)
- Dictionaries (mutable, unordered)
- Set (immutable, unordered)

Functions and Modular Programming

100 mins

Functions are blocks of reusable code that can be called multiple times from different parts of a program. Modular programming is the practice of organizing a program into separate modules that can be imported and used as needed.

Key Topics:

- Function definition and syntax
- Function arguments and return values
- Scope and namespaces
- Importing and exporting modules

Error Handling and Exception Management

100 mins

Errors are unexpected events that occur during the execution of a program. Exception management is the process of handling errors in a controlled and graceful manner.

Key Topics:

- Exception types and hierarchy
- Try, except, and finally blocks
- Custom exception classes
- Logging and debugging

Object-Oriented Programming (OOP) in Python

100 mins

Object-oriented programming (OOP) is a programming paradigm that uses objects and classes to represent and manipulate data. Python is a multi-paradigm language that supports OOP.

Key Topics:

- Classes and objects
- Attributes and methods
- Inheritance and polymorphism
- Encapsulation and abstraction

Introduction to Python Libraries and Modules

100 mins

Python has a vast ecosystem of libraries and modules that can be used to extend the functionality of the language. These libraries and modules are organized into packages and can be installed using package managers like pip.

Key Topics:

- Package management (pip, conda)
- Common Python libraries (NumPy, Pandas, Matplotlib)
- Module structure and organization

Final Project and Code Best Practices

100 mins

The final project is a practical application of the concepts learned in the previous units. It involves designing, implementing, and testing a small program that solves a specific problem. Code best practices are guidelines that help developers write clean, readable, and maintainable code.

Key Topics:

- Project planning and design
- Code organization and structure
- Testing and debugging
- Documentation and comments

What is Python? History, Syntax, and Environment

10 mins

Python is a high-level, interpreted, general-purpose programming language. It was created by Guido van Rossum in the late 1980s and is named after the British comedy group Monty Python. Python's syntax is designed to be clear and readable, and it has a large and active community of developers.

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Installing Python and Setting Up the Environment

10 mins

Before you can start writing Python code, you need to install the language and set up your development environment. This includes installing Python, a code editor, and a virtual environment.

Key Topics:

- Installing Python
- Setting up a code editor
- Creating a virtual environment

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